

ROADEYE

V 1.0

RIDE EBYOND YOUR VISION

Group 15

USER MANUAL

Helmet Unit • Bike Module • Mobile Application

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1. Introduction

RoadEye is a smart motorcycle safety system designed to combine a motorcycle-mounted sensing unit, a helmet-based user interface, and a mobile application into a single rider-support system. The purpose of the system is to provide safety-related information and assistance without requiring the rider to continuously look down at a mobile phone.

1.1 Purpose of this Manual

This manual explains how to set up, connect, operate, maintain, and troubleshoot the RoadEye system. It is written so that a first time user can follow the setup and feature instructions without needing detailed software or electronics knowledge.

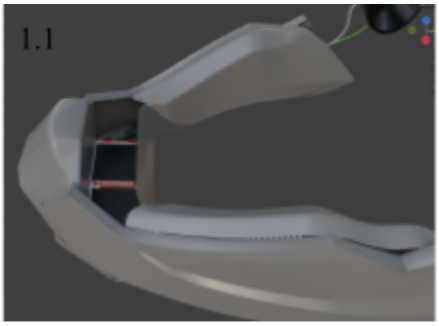
1.2 Scope

- Physical setup of the RoadEye helmet unit and bike module.
- Mobile application installation and initial configuration.
- Pairing and connecting the system components.
- Use of the documented safety, navigation, emergency, analytics, and related functions that are confirmed as implemented.
- Basic maintenance and troubleshooting.

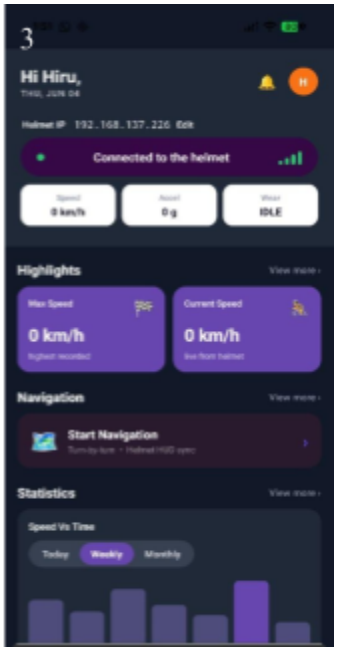
1.3 Before You Start

Confirm that you have the complete hardware, a compatible phone, the required installation package/application access, and the latest project build. Do not treat an unfinished prototype function as a production safety guarantee.









1 - Helmet Module

2 - Bike Module

3 - Mobile Application

2. System Overview

RoadEye is organized into three main user-facing parts: the helmet unit, the bike module, and the mobile application. These components work together to collect information, communicate with the rider, and support safety-related functions.

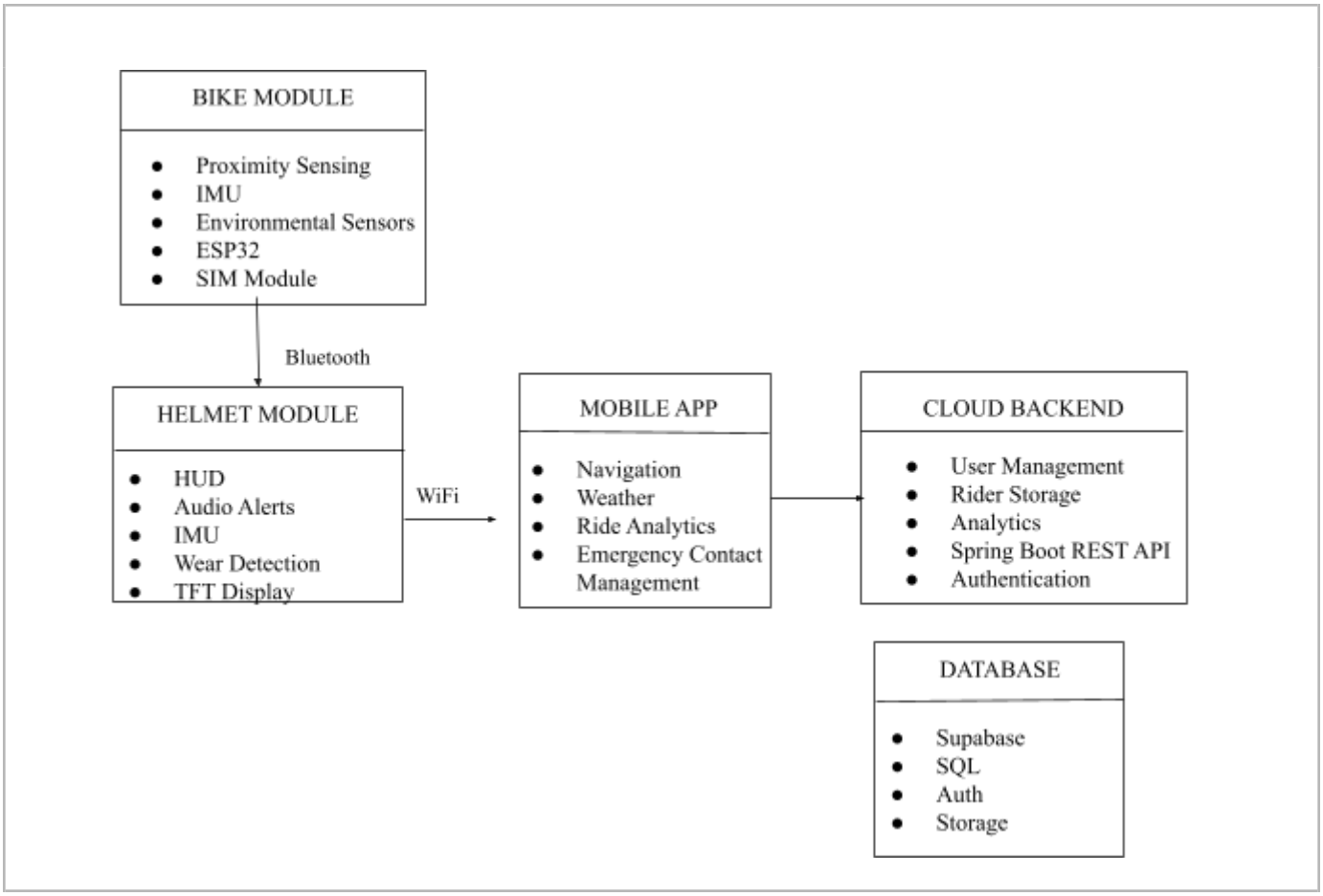
2.1 Main Modules

Module	Purpose
Helmet Unit	Provides the rider-facing interface and safety feedback through the helmet-mounted display/HUD and other user feedback components described in the project.
Bike Module	Mounted on the motorcycle and provides sensing/vehicle-side data used by the RoadEye system.
Mobile Application	Provides user interaction, configuration, navigation, ride information, emergency-related functions, and other application-level features.



2.2 Typical Operating Flow

A typical ride begins with powering the hardware, opening the application, connecting the system components, confirming the required permissions and emergency information, and then starting the ride. During the ride, the system can provide the implemented safety and information features through the appropriate interface.



3. User Roles and Intended Audience

There are four possible audiences. This manual primarily targets end users while also providing enough setup information for installers/technicians and enough system context for evaluators.

Audience	What they need from this manual
End users (Riders)	Simple setup, pairing, operating instructions, safety precautions, maintenance, and troubleshooting.
Admin / moderator users	Configuration and support information where applicable to the project build.
Installers / technicians	Hardware placement, connection, basic wiring/assembly guidance, and component identification.

4. Safety Precautions

RoadEye is a rider-assistance prototype/system. It must not be treated as a replacement for safe riding, visual awareness, traffic rules, or emergency services.

4.1 General Riding Safety

- Always wear and correctly fasten an approved motorcycle helmet.
- Keep your attention on the road and surrounding traffic.
- Do not operate the mobile phone manually while riding.
- Do not rely solely on RoadEye alerts or navigation.
- Stop in a safe location before changing settings or troubleshooting the system.

4.2 Hardware Safety

- Do not expose electronic modules to water unless the enclosure has been specifically verified for water resistance.
- Do not open, modify, or short-circuit batteries.
- Inspect mounting and cables before a ride.
- Do not position sensors or the display where they obstruct the rider's normal field of view.

4.3 Emergency Feature Limitation

Emergency detection and notification features should be treated as assistance mechanisms. Network availability, sensor conditions, phone permissions, and hardware state can affect their operation.

5. Hardware Setup

Complete the physical setup before attempting full system operation.

5.1 Identify the Hardware



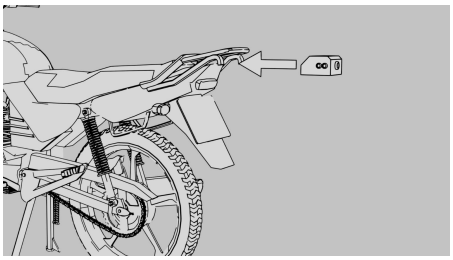
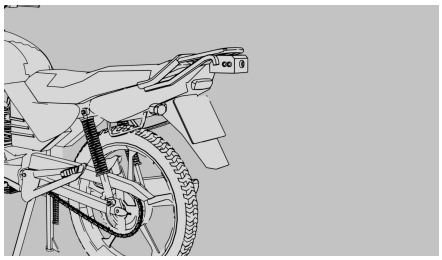
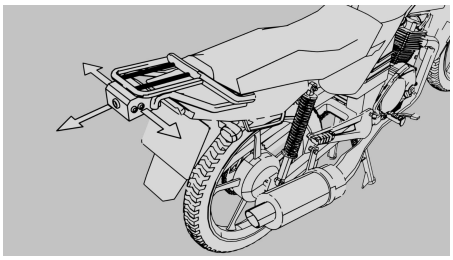






5.2 Bike Module Installation

Mount the bike module in the position at the back of the motor cycle (center).

Ensure the module is firmly secured and that its sensing elements have a clear field of view.

Installation steps:

1. Turn off the motorcycle and RoadEye hardware before installation.
2. Position the mounting module at the designated location.
3. Secure the module so that it does not move during normal riding.
4. Confirm that sensors are unobstructed.
5. Check cable routing and ensure cables cannot interfere with steering, wheels, brakes, or moving parts.
6. Power on the module and verify its status indication.

5.3 Helmet Unit Setup

7. Confirm that the display/HUD is aligned with the rider's normal line of sight.
8. Check the buckle/wear-related sensing components if present in the final build.
9. Power on the unit and verify its status indication.
10. Confirm that the helmet and the mobile app are connected.

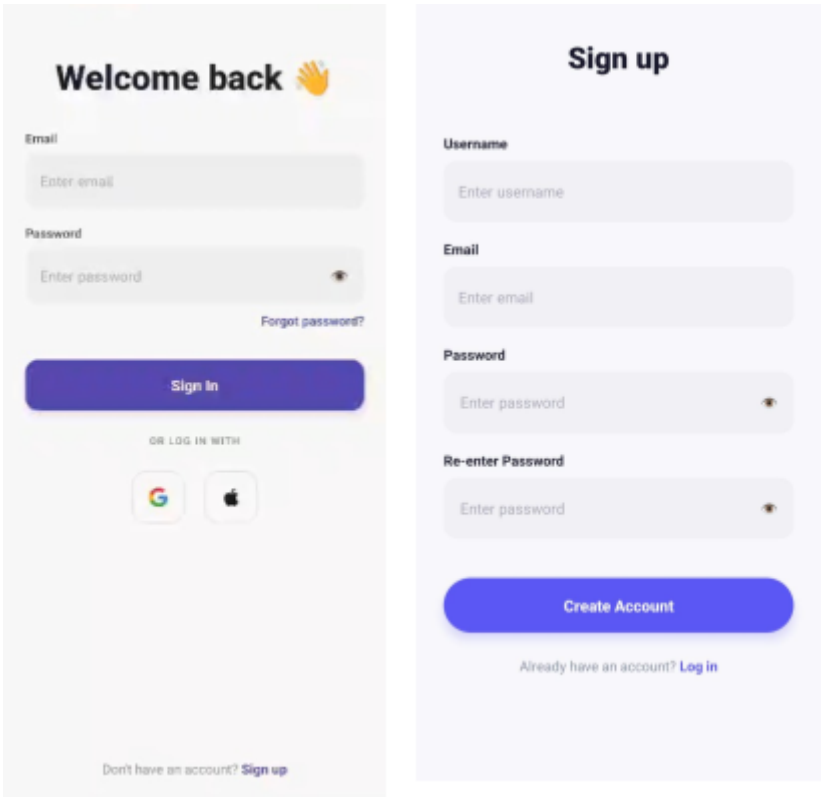
6. Software Setup

6.1 Mobile Application Installation

Install the RoadEye application using the distribution method provided by the project team. If the application is not publicly released, use the official project installation package or QR code supplied for the demonstration build. (we will provide the QR soon)

11. Install the RoadEye application.
12. Open the application.

13. Create an account or sign in if authentication is enabled in the current build.
14. Allow only the permissions required by the implemented features.
15. Complete the initial configuration.



6.2 Required Permissions

Permission / Access	Why it may be required
Location	Navigation, positioning, ride tracking or emergency location features.
Bluetooth	Communication with supported nearby devices, if used by the final build.
Wi-Fi / Network	Communication with online services or connected devices, if used.
Notifications	Safety alerts and system notifications.
Other	Only document permissions actually requested by the application.

6.3 Initial Configuration

- Configure the user profile if required.
- Add emergency contacts before relying on emergency features.
- Pair the required RoadEye hardware.
- Confirm that the dashboard shows the expected connection/device status.



7. Getting Started

7.1 Quick Start

1. Charge the RoadEye hardware.
2. Install/open the mobile application.
3. Power on the helmet unit and bike module.
4. Pair/connect the devices.
5. Configure emergency contacts and permissions.
6. Check the dashboard/device status.
7. Start the ride only after confirming that the system is ready.

8. Feature-by-Feature Instructions

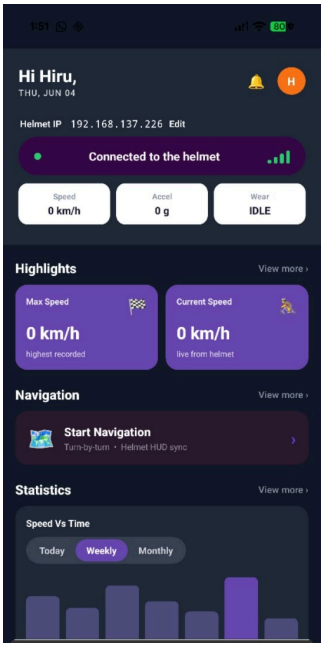
The following feature sections are written as operating guides.

1. 8.1 Dashboard / Home Screen – How to open and use the dashboard.
2. 8.2 Device Pairing and Connection – How to connect the bike/helmet modules.
3. 8.3 Navigation – How to enter a destination and start navigation.
4. 8.4 Ride Tracking and Analytics – How to start a ride and view statistics.
5. 8.5 Safety / Nearby-Object Alerts – How proximity alerts work and what the rider sees/hears.
6. 8.6 Crash Detection / Emergency Workflow – What happens when a crash is detected.
7. 8.7 Emergency Contacts – How to add and manage emergency contacts.
8. 8.8 Weather / Environmental Information – How to view environmental information.

9. 8.9 Music / Audio Functions – How to use music/audio controls.
10. 8.10 Theft Detection – State that theft detection is activated and that a detected theft triggers an alert in the connected mobile app. Your current manual already says this.
11. 8.11 Authentication and Account Management – How to register, log in, and manage the account.

8.1 Dashboard / Home Screen

Open the application after signing in. The dashboard provides the main entry point to the RoadEye functions and displays the available system/ride information in the current build.



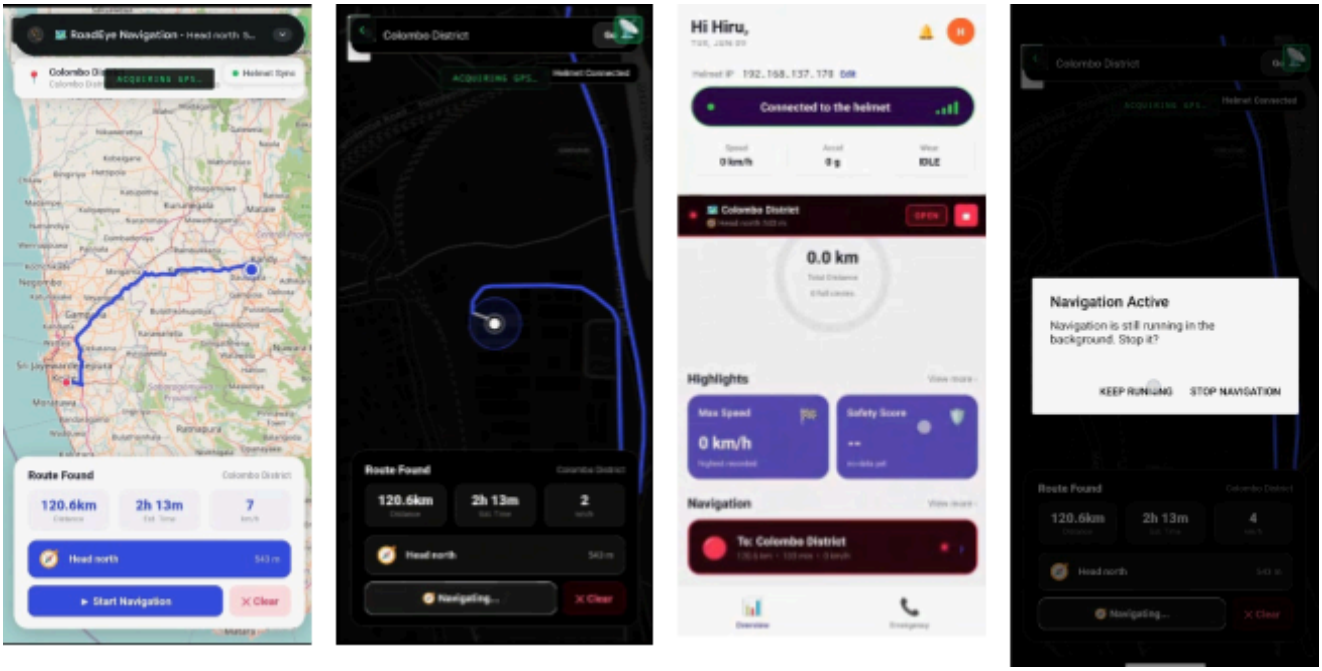
8.2 Device Pairing and Connection

Open the device/pairing section, power on the relevant RoadEye hardware, select the detected device, and confirm the connection. Verify that the application indicates the expected connected state.



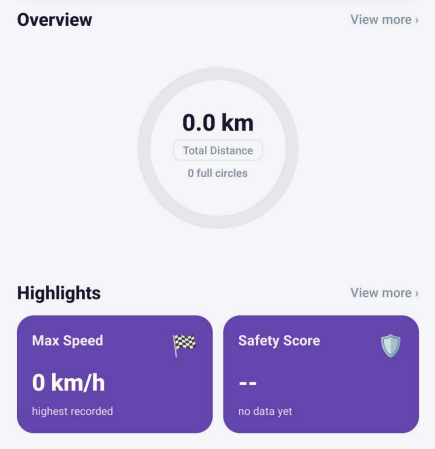
8.3 Navigation

Open the navigation function, enter or select a destination, review the route, and start navigation. Where the final build supports HUD guidance, the rider-facing display can present navigation information without requiring the rider to continuously look at the phone.



8.4 Ride Tracking and Analytics

Start a ride/session using the application's ride function. At the end of the ride, open ride history/analytics to review the information recorded by the current implementation, such as trip information and available ride statistics.



8.5 Safety / Nearby-Object Alerts

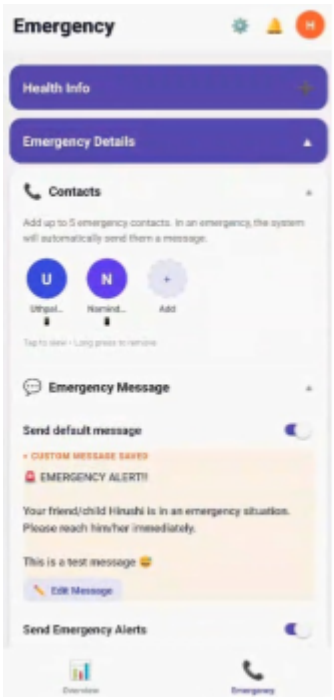
When the sensing system detects a configured condition, the system provides the feedback implemented in the current build. The exact alert type and threshold must be documented using the team's verified test configuration.

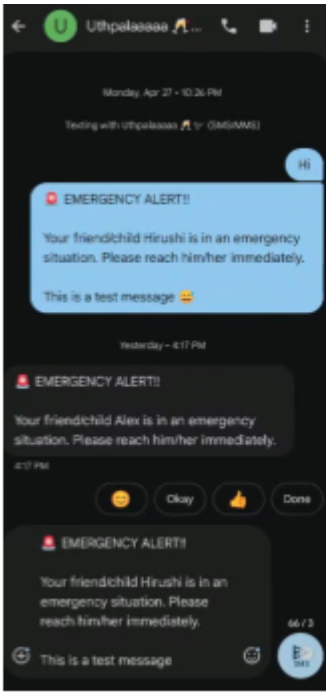
8.6 Crash Detection / Emergency Workflow

The implemented emergency workflow should be tested before demonstration. When a crash condition is detected, the system should follow the configured workflow, including any confirmation and emergency-contact notification that is actually implemented.

8.7 Emergency Contacts

Open the emergency-contact settings, add the required contact information, save the contact, and verify that the information is displayed correctly. Test the complete notification path before relying on it.





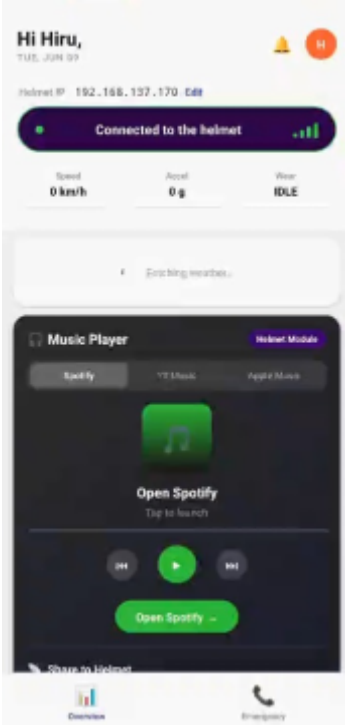
8.8 Weather / Environmental Information

Open the environmental/weather section if enabled in the current build. Review the information displayed and verify that network-dependent data is updating correctly.



8.9 Music / Audio Functions

If the current build provides music/audio controls, use the media controls provided in the application and verify that audio output is routed to the supported helmet audio interface.

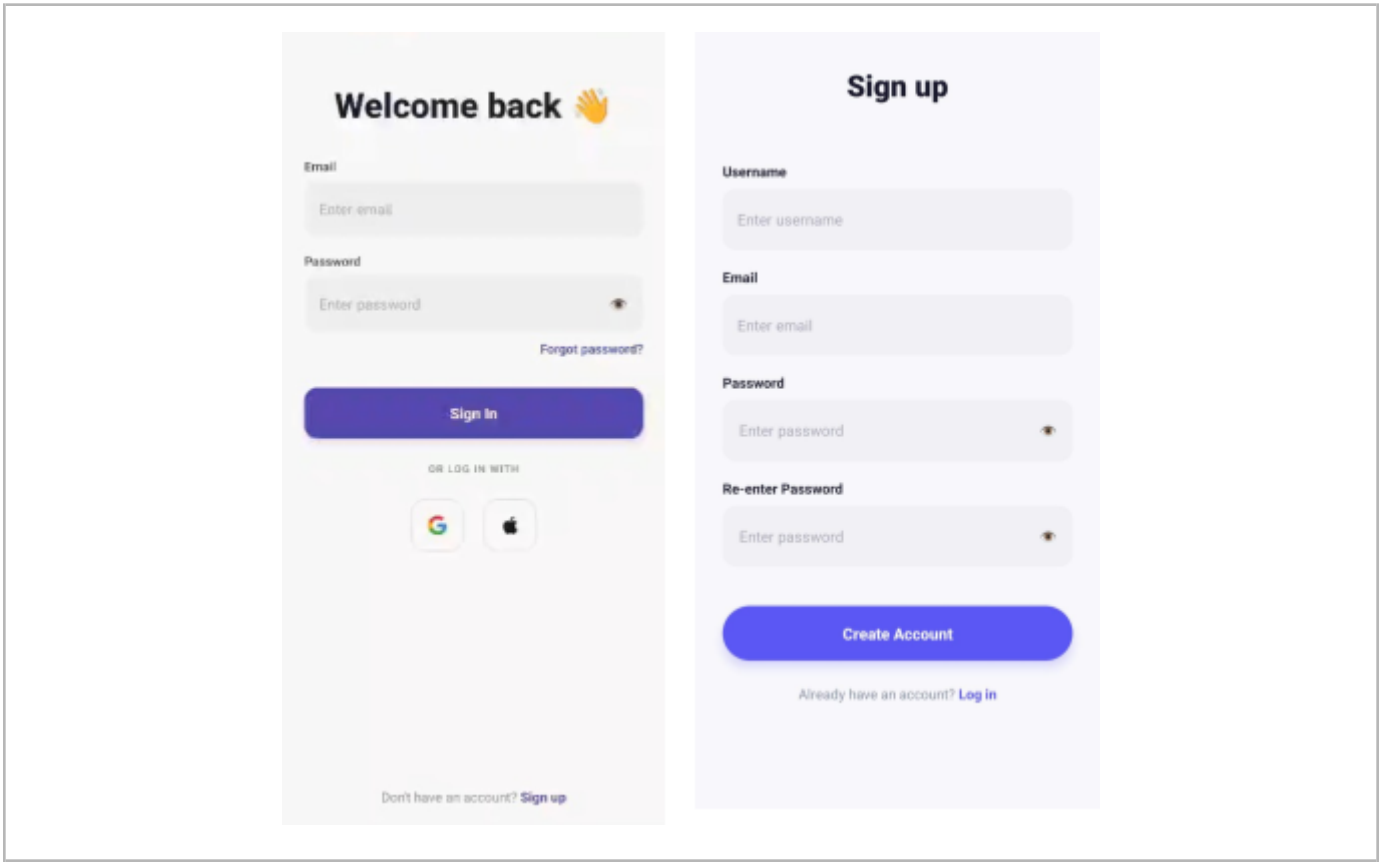


8.10 Theft Detection

Theft detection is activated. When a potential theft event is detected, the system automatically sends an alert to the connected mobile application.

8.11 Authentication and Account Management

Use the login/registration functions provided by the current application build. Keep account credentials private and sign out when using a shared test device.



9. Maintenance and Care

9.1 Before Every Ride

- Check helmet and bike module mounting.
- Check battery/charging status.
- Check that sensors are not blocked.
- Check that the HUD/display is positioned correctly.
- Check that the mobile application and hardware are connected.

9.2 Cleaning

- Use a soft, dry or lightly damp cloth appropriate for the external enclosure.
- Keep sensor surfaces and optical/display surfaces clean.
- Do not spray liquid directly onto electronic modules.
- Do not disassemble the electronics for routine cleaning.

9.3 Storage

- Store the electronic units in a dry, safe location.
- Avoid excessive heat and direct exposure to harsh environmental conditions.
- Charge/store batteries according to the actual battery and charging design used by the project.

10. Troubleshooting / FAQ

Problem	Possible cause	Recommended action
Application cannot connect to hardware	Hardware off, pairing issue, permissions, or communication unavailable.	Power-cycle the relevant devices, verify permissions, and retry pairing.
No dashboard/device status	Application not connected or service unavailable.	Check network/device state and restart the application if necessary.
Navigation does not work	Location permission/network unavailable or feature not enabled.	Check location permission and network access; verify the final build supports navigation.
No safety alert during a test	Sensor alignment, threshold/configuration, or hardware state.	Check sensor placement and run the approved test procedure.
Emergency notification not received	No network, incorrect contact details, permissions, or incomplete workflow.	Verify contact details, phone/network state, and test the complete emergency path.
HUD/display not visible	Power, alignment, display connection, or environmental conditions.	Check power and physical alignment; do not adjust while riding.
Audio not heard	Volume, connection, or speaker hardware issue.	Check connection and volume; test audio while stationary.
Device battery is low	Insufficient charging or high power use.	Charge the device using the project's approved charging method.
Application crashes	Software/build issue or unsupported device state.	Restart the application and record the error for the project team.

10.1 Frequently Asked Questions

Q: Can I use RoadEye without the mobile application?
A: Use only the operating modes confirmed by the final project build. Do not assume that every feature is available independently.

Q: Should I interact with the phone while riding?
A: No. Configure the system before riding and stop safely before making changes.

Q: Are RoadEye alerts guaranteed to detect every hazard or crash?
A: No. Treat all alerts as assistance and maintain normal rider awareness.

11. Contact / Support Information

Support item	Details to insert
Email	roadeye3yp@gmail.com
Website	https://magnetic-ride-vision-pro.base44.app/